MSML610: Advanced Machine Learning

Lesson 05.2: Overfitting

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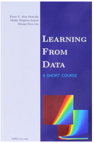


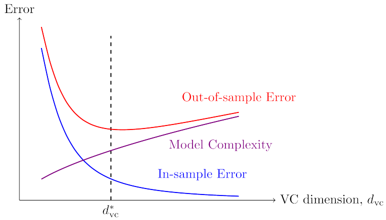
References:

- Abu-Mostafa et al.: *"Learning From Data"* (2012)

- ***Overfitting***
- Bias Variance Analysis
- Learning Curves

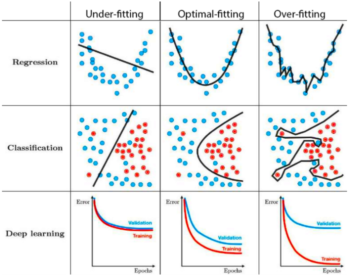
Overfitting: Definition

- **Overfitting** occurs when the model fits the data more than what is warranted
- Surpass point where E_{out} is minimal (optimal fit)
  - Model complexity too high for data/noise
  - Noise in training set mistaken for signal
- **Fitting noise instead of signal** is not useless but harmful
  - Model infers in-sample pattern that, when extrapolated out-of-sample, deviates from target function $\implies$ poor generalization



Optimal Fit

- The opposite of overfitting is **optimal fit**
  - Train a model with proper complexity for the data
- The optimal fit:
  - Implies E_{out} is minimal
  - Does not imply generalization error $E_{out} - E_{in}$ is minimal
    - E.g., no training implies generalization error equal to 0
- The **generalization error** is the additional error $E_{out} - E_{in}$ when going from in-sample to out-of-sample



Overfitting: Diamond Price Example

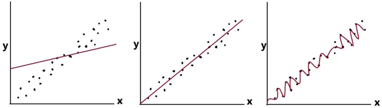
- Predict diamond price as a function of carat size (regression problem)
- **True relationship**

$$\text{price} \sim (\text{carat size})^2 + \varepsilon$$

  where:

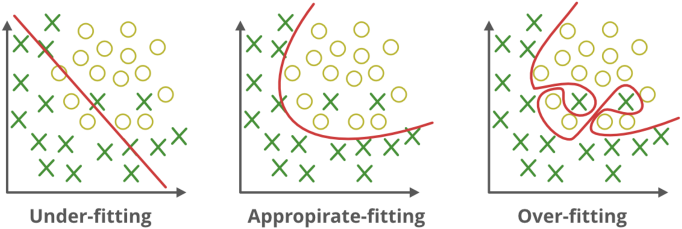
  – Square function: price increases more with rarity
  – Noise ε: e.g., market noise, missing features

- **Fit with:**
  – *Line*
    ∗ Underfit
    ∗ High bias (large error)
    ∗ Low variance (stable model)
  – *Polynomial of degree 2*
    ∗ right fit
  – *Polynomial of degree 10*
    ∗ Overfit (wiggly curve)
    ∗ Low bias
    ∗ High variance (many



Overfitting: Classification Example

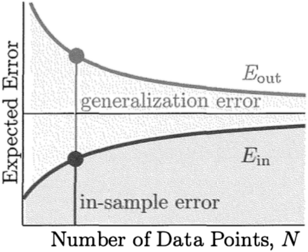
- Assume:
  - You want to separate 2 classes using 2 features x_1, x_2
  - The true class boundary has a parabola shape
- You can use logistic regression and a decision boundary equal to:
  - A line $\text{logit}(w_0 + w_1 x + w_2 y)$
    - Underfit
    - High bias, low variance
  - A parabola $\text{logit}(w_0 + w_1 x + w_2 x^2 + w_3 xy + w_4 y^2)$
    - Right fit
  - A wiggly decision boundary $\text{logit}(w_0 + \text{high powers of } x_1, x_2)$
    - Overfit
    - Low bias, high variance



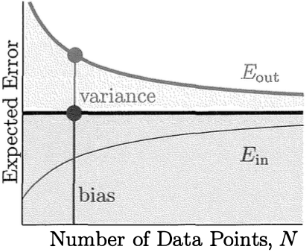
- Overfitting
- ***Bias Variance Analysis***
- Learning Curves

VC Analysis vs Bias-Variance Analysis

- Both VC analysis and bias-variance analysis are concerned with the hypothesis set $\mathcal{H}$
  - **VC analysis**:
    $$E_{out} \leq E_{in} + \Omega(\mathcal{H})$$
  - **Bias-variance analysis**
    $$E_{out} = \text{bias} + \text{variance}$$



VC Analysis



Bias-Variance Analysis

Hypothesis Set and Bias-Variance Analysis

- **Learning** consists in finding $g \in \mathcal{H}$ such that $g \approx f$ where f is an unknown function
- The **tradeoff in learning** is between:
  - Bias vs variance
  - Overfitting vs underfitting
  - More complex vs less complex $\mathcal{H}$ / h
  - Approximation (in-sample) vs generalization (out-of-sample)

Decomposing Error in Bias-Variance (1/4)

- **Problem**
  - Regression set-up: target is a real-valued function
  - Hypothesis set $\mathcal{H} = \{h_1(\underline{\boldsymbol{x}}), h_2(\underline{\boldsymbol{x}}), ...h_n(\underline{\boldsymbol{x}})\}$
  - Training data $\mathcal{D}$ with N examples
  - Squared error $E_out = \mathbb{E}[(g(\underline{\boldsymbol{x}}) - f(\underline{\boldsymbol{x}}))^2]$
  - Choose the best function $g \in \mathcal{H}$ that approximates unknown f
- **Question**
  - What is the out-of-sample error $E_{out}(g)$ as function of $\mathcal{H}$ for a training set of N examples?

Decomposing Error in Bias-Variance (2/4)

- The final hypothesis g depends on training set D, so make the dependency explicit $g^{(D)}$:

$$E_{out}(g^{(D)}) \triangleq \mathbb{E}_{\underline{\boldsymbol{x}}}[(g^{(D)}(\underline{\boldsymbol{x}}) - f(\underline{\boldsymbol{x}}))^2]$$

- Interested in:
  - Hypothesis set $\mathcal{H}$ rather than specific h
  - Training set D of N examples, rather than a specific D
- Remove dependency from D by averaging over all possible training sets D with N examples:

$$E_{out}(\mathcal{H}) \triangleq \mathbb{E}_D[E_{out}(g^{(D)})] = \mathbb{E}_D[\mathbb{E}_{\underline{\boldsymbol{x}}}[(g^{(D)}(\underline{\boldsymbol{x}}) - f(\underline{\boldsymbol{x}}))^2]]$$

Decomposing Error in Bias-Variance (3/4)

- Switch the order of the expectations since the quantity is non-negative:

$$E_{out}(\mathcal{H}) = \mathbb{E}_{\underline{\boldsymbol{x}}}[\mathbb{E}_D[(g^{(D)}(\underline{\boldsymbol{x}}) - f(\underline{\boldsymbol{x}}))^2]$$

- Focus on $\mathbb{E}_D[(g^{(D)}(\underline{\boldsymbol{x}}) - f(\underline{\boldsymbol{x}}))^2]$ which is a function of $\underline{\boldsymbol{x}}$
- Define the *average hypothesis* over all training sets as:

$$\overline{g}(\underline{\boldsymbol{x}}) \triangleq \mathbb{E}_D[g^{(D)}(\underline{\boldsymbol{x}})]$$

- Add and subtract it inside the $\mathbb{E}_D$ expression:

$$\begin{aligned}
E_{out}(\mathcal{H}) =& \mathbb{E}_{\underline{\boldsymbol{x}}}\left[\mathbb{E}_D\left[\left(g^{(D)}(\underline{\boldsymbol{x}}) - f(\underline{\boldsymbol{x}})\right)^2\right]\right] \\
=& \mathbb{E}_{\underline{\boldsymbol{x}}}\mathbb{E}_D[(g^{(D)} - \overline{g} + \overline{g} - f)^2] \\
=& \mathbb{E}_{\underline{\boldsymbol{x}}}\mathbb{E}_D[(g^{(D)} - \overline{g})^2 + (\overline{g} - f)^2 + 2(g^{(D)} - \overline{g})(\overline{g} - f)] \\
& (\mathbb{E}_D \text{ is linear and } (\overline{g} - f) \text{ doesn't depend on } D) \\
=& \mathbb{E}_{\underline{\boldsymbol{x}}}\left[\mathbb{E}_D[(g^{(D)} - \overline{g})^2] + (\overline{g} - f)^2 + 2\mathbb{E}_D[(g^{(D)} - \overline{g})](\overline{g} - f)\right]
\end{aligned}$$

Decomposing Error in Bias-Variance (4/4)

- The cross term:

$$\mathbb{E}_D[(g^{(D)} - \overline{g})](\overline{g} - f)$$

disappears since applying the expectation on D, it is equal to:

$$(g^{(D)} - \mathbb{E}_D[\overline{g}])(\overline{g} - f) = 0 \cdot (\overline{g} - f) = 0 \cdot \text{constant}$$

- Finally:

$$\begin{aligned} E_{out}(\mathcal{H}) &= \mathbb{E}_{\underline{\boldsymbol{x}}}[\mathbb{E}_D[(g^{(D)} - \overline{g})^2] + (\overline{g}(\underline{\boldsymbol{x}}) - f(\underline{\boldsymbol{x}}))^2] \\ &= \mathbb{E}_{\underline{\boldsymbol{x}}}[\mathbb{E}_D[(g^{(D)} - \overline{g})^2]] + \mathbb{E}_{\underline{\boldsymbol{x}}}[(\overline{g} - f)^2] \quad (\mathbb{E}_{\underline{\boldsymbol{x}}} \text{ is linear}) \\ &= \mathbb{E}_{\underline{\boldsymbol{x}}}[\text{var}(\underline{\boldsymbol{x}})] + \mathbb{E}_{\underline{\boldsymbol{x}}}[\text{bias}(\underline{\boldsymbol{x}})^2] \\ &= \text{variance} + \text{bias} \end{aligned}$$

Interpretation of Average Hypothesis

- The **average hypothesis** over all training sets

$$\overline{g}(\underline{\boldsymbol{x}}) \triangleq \mathbb{E}_D[g^{(D)}(\underline{\boldsymbol{x}})]$$

  can be interpreted as the "best" hypothesis from $\mathcal{H}$ training on N samples
  - Note: $\overline{g}$ is not necessarily $\in \mathcal{H}$
- In fact it's like **ensemble learning**:
  - Consider all the possible data sets D with N samples
  - Learn g from each D
  - Average all the hypotheses

Interpretation of Variance and Bias Terms

- The out-of-sample error can be decomposed as:

$$E_{out}(\mathcal{H}) = \text{bias}^2 + \text{variance}$$

- **Bias term**

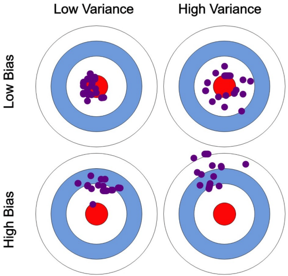
$$\text{bias}^2 = \mathbb{E}_{\underline{\boldsymbol{x}}}[(\overline{g}(\underline{\boldsymbol{x}}) - f(\underline{\boldsymbol{x}}))^2]$$

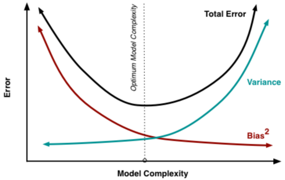
  - Does not depend on learning as it is not a function of the data set D
  - Measures how limited $\mathcal{H}$ is
    - I.e., the ability of $\mathcal{H}$ to approximate the target with infinite training sets

- **Variance term**

$$\text{variance} = \mathbb{E}_{\underline{\boldsymbol{x}}}\mathbb{E}_D[(g^{(D)}(\underline{\boldsymbol{x}}) - \overline{g}(\underline{\boldsymbol{x}}))^2]$$

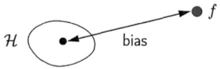
  - Measures variability of the learned hypothesis from D for any $\underline{\boldsymbol{x}}$
    - With infinite training sets, we could focus on the "best" g



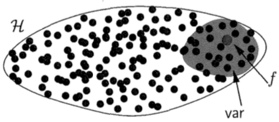


Variance and Bias Term Varying Cardinality of $\mathcal{H}$

- If hypothesis set **has a single function**:
  $\mathcal{H} = \{h \neq f\}$
  - Large bias
    - h might be far from f
  - Variance $= 0$
    - No cost in choosing hypothesis



- If hypothesis set **has many functions**:
  $\mathcal{H} = \{\text{many hypotheses } h\}$
  - Bias can be 0
    - E.g., if $f \in \mathcal{H}$
  - Large variance
    - Depending on data set D, end up far from f
    - Larger $\mathcal{H}$, farther g from f

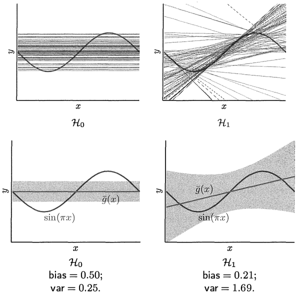


Bias-Variance Trade-Off: Numerical Example

- **Machine learning problem**:
  - Target function $f(x) = \sin(\pi x), x \in [-1, 1]$
  - Noiseless target
  - You have $f(\underline{\boldsymbol{x}})$ for $N = 2$ points
- **Two hypotheses sets $\mathcal{H}$**:
  - Constant model: $\mathcal{H}_0 : h(x) = b$
  - Linear model: $\mathcal{H}_1 : h(x) = ax + b$
- **Which model is best**?
  - Depends on the perspective!
  - Best for *approximation*: minimal error approximating the sinusoid
  - Best for *learning*: learn the unknown function with minimal error from 2 points

Bias-Variance Trade-Off: Numerical Example

- **Approximation**
  - $E_{out}(g_0) = 0.5$
    - g_0 is a constant and approximates the sinusoid poorly (higher bias)
  - $E_{out}(g_1) = 0.2$
    - g_1 is a line and has more degrees of freedom (lower bias)
  - The line model *approximates better* than the constant model



- **Learning**
  - Algorithm:
    - Pick 2 points as training set D
    - Learn g from D
    - Compute $\mathbb{E}_D[E_{out}(g)]$
  - Average over all data sets D:

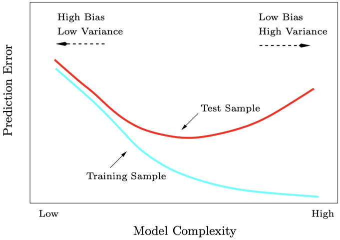
$$E_{out} = \text{bias}^2 + \text{variance}$$

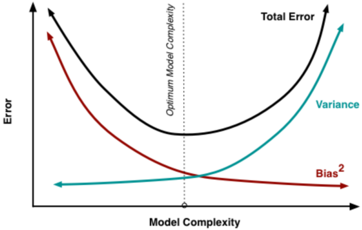
  - $E_{out}(g_0) = 0.5 + 0.25 = 0.75$
    - g_0 is more stable given the data set (lower variance)
  - $E_{out}(g_1) = 0.2 + 1.69 = 1.9$

Bias-Variance Curves

- **Bias-variance curves** are plots of E_{out} increasing the complexity of the model

- Typical **shape** of bias-variance curves
  - E_{in} and E_{out} start from the same point
  - E_{in}
    - ∗ Is decreasing with increasing model complexity
    - ∗ Can even go to 0
  - E_{out}
    - ∗ Is always larger than E_{in}
    - ∗ Is the sum of bias and variance
    - ∗ Has a bowl shape
    - ∗ Reaches a minimum for optimal fit
    - ∗ Before the minimum



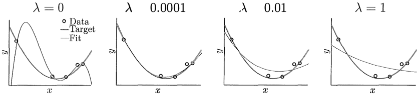


Bias-Variance Curves and Regularization

- Use **model with regularization** to learn at the same time:
  - Model coefficients $\underline{\boldsymbol{w}}$
  - Model "complexity" (e.g., VC dimension)
- Learn the optimal model $\underline{\boldsymbol{w}}(\lambda)$ as a function of $\lambda = \{..., 10^{-1}, 1.0, 10, ...\}$ by optimizing:

$$\underline{\boldsymbol{w}}(\lambda) = \text{argmin}_{\underline{\boldsymbol{w}}} E_{aug}(\underline{\boldsymbol{w}}) = E_{in}(\underline{\boldsymbol{w}}) + \Omega(\lambda)$$

- Interpret regularization parameter:
  - Small λ:
    - ∗ Complex model
    - ∗ Low bias
    - ∗ High variance
  - Large λ:
    - ∗ Simple model
    - ∗ High bias
    - ∗ Low variance
  - An intermediate λ is optimal



How to Measure the Model Complexity

- Number of features
- Parameters for model form / degrees of freedom, e.g.,
  - VC dimension d_{VC}
  - Degree of polynomials
  - k in KNN
  - ν in NuSVM
- Regularization param λ
- Training epochs for neural network

Bias-Variance Decomposition with a Noisy Target

- Extend bias-variance decomposition to **a noisy target**

$$y = f(\underline{\boldsymbol{x}}; \underline{\boldsymbol{w}}) + \varepsilon = \underline{\boldsymbol{w}}^T \underline{\boldsymbol{x}} + \varepsilon$$

- With similar hypothesis and analysis conclude that:

$$\begin{aligned} E_{out}(\mathcal{H}) &= \mathbb{E}_{D,\underline{\boldsymbol{x}}}\left[(g^{(D)} - \overline{g})^2\right] + \mathbb{E}_{\underline{\boldsymbol{x}}}\left[(\overline{g} - f)^2\right] + \mathbb{E}_{\varepsilon,\underline{\boldsymbol{x}}}\left[(f - y)^2\right] \\ &= \text{variance} + \text{bias} + \text{stochastic noise} \end{aligned}$$

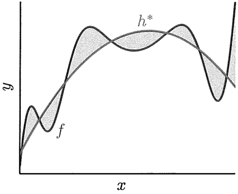
- The out-of-sample error is the sum of 3 contributions
  1. **Variance**: from the set of hypotheses to the centroid of the hypothesis set
  2. **Bias**: from the centroid of the hypothesis set to the noiseless function
  3. **Noise**: from the noiseless function to the real function

Bias as Deterministic Noise

- The bias term can be interpreted as **deterministic noise**
- **Bias** is the part of the target function that hypothesis set cannot capture:

$$h^*(\underline{\boldsymbol{x}}) - f(\underline{\boldsymbol{x}})$$

  where:
  - $h^*()$ is the best approximation of $f(\underline{\boldsymbol{x}})$ in the hypothesis set $\mathcal{H}$ (e.g., $\overline{g}(x)$)
- The hypothesis set $\mathcal{H}$ cannot learn the deterministic noise since it is outside of its ability, and thus it behaves like "noise"

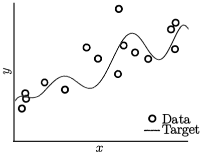


Deterministic vs Stochastic Noise in Practice

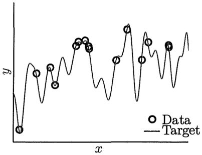
- **Deterministic noise:**
  - Fixed for a particular $\underline{\boldsymbol{x}}$
  - Depends on $\mathcal{H}$
  - Independent of ε or D
- **Stochastic noise:**
  - Not fixed for $\underline{\boldsymbol{x}}$
  - Independent of D or $\mathcal{H}$
- In a machine learning problem, no difference exists between stochastic and deterministic noise, as $\mathcal{H}$ and D are fixed
  - From the training set alone, you cannot determine if data is from a *noiseless complex* target or a *noisy simple* target

Deterministic vs Stochastic Noise Example

- **Two targets**:
  - Noisy low-order target (10th order polynomial)
  - Noiseless high-order target (50th order polynomial)
- **Training set**
  - Generate $N = 15$ data points
- **Two models**
  - $\mathcal{H}_2$ low-order hypothesis (2nd order polynomial)
  - $\mathcal{H}_{10}$ high-order hypothesis (10th order polynomial)
- Learning model:
  - Sees only training samples
  - Can't distinguish noise sources



(a) 10th order target function



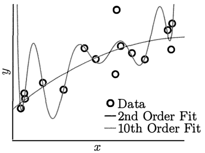
(b) 50th order target function

Deterministic vs Stochastic Noise Example

- Noisy low-order target:
  - Fit 2nd to 10th order polynomial
  - $\downarrow E_{in}$ (more degrees of freedom), $\uparrow\uparrow E_{out}$ (fits noise)
- Noiseless high-order target
  - Fit 2nd to 10th order polynomial
  - Same phenomenon
  - $\downarrow E_{in}$ (more degrees of freedom), $\uparrow\uparrow E_{out}$ (fits noise)
- **Wrong approach**
  - Target is 10th order polynomial
  - Use 10-order hypothesis to fit target perfectly
- **Right approach**
  - Consider number of data points, i.e., 15 points
  - Rule of thumb from VC analysis

    $$\text{degrees of freedom of model} = \text{number of data points} / 10$$

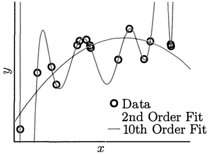
  - Use 1 or 2 degrees of freedom



(a) Noisy low order target

10th order noisy target

	2nd Order	10th Order
E_{in}	0.050	0.034
E_{out}	0.127	**9.00**



(b) Noiseless high order target

50th order noiseless target

	2nd Order	10th Order
E_{in}	0.029	10^{-5}
E_{out}	0.120	**7680**

Amount of Data and Model Complexity

- One must match the *model complexity* to
  - **Data resources**
  - **Signal to noise ratio**
  - **Not target complexity**
- The rule of thumb is:

$$d_{VC}(\text{degrees of freedom of the model}) = N(\text{number of data points})/10$$

  - In other words, 10 data points needed to fit a degree of freedom
  - If the data is noisy, you need even more data

Overfitting as a Function of Data Resources, Model Complexity, Noise

- You can measure overfitting in terms of **generalization error** $\frac{E_{out} - E_{in}}{E_{out}}$
  - $\uparrow$ data resources $N \implies \downarrow$ overfitting
  - $\uparrow$ model complexity $d_{VC} \implies \uparrow$ overfitting
  - $\uparrow$ deterministic noise (target complexity) $\implies \uparrow$ overfitting
  - $\uparrow$ stochastic noise $\sigma^2 \implies \uparrow$ overfitting

- Overfitting
- Bias Variance Analysis
- ***Learning Curves***

Learning Curves vs Bias-Variance Curves

- **Learning curves** are the dual of the bias-variance curves
- For **bias-variance curves**

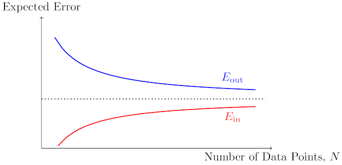
$$E_{in}, E_{out} = f(d_{VC}|N)$$

  - Keep the complexity of training set fixed (number of examples N)
  - Vary the model in terms of:
    * Model complexity d
    * Number of features p
    * Regularization amount λ

- For **learning curves**

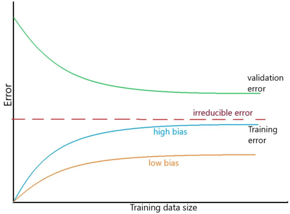
$$E_{in}, E_{out} = f(N|d_{VC})$$

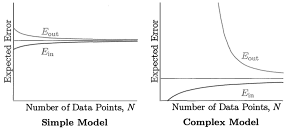
  - Fix the model
  - Vary the size N of training set



Typical Form of Learning Curves

- Learning curves plot E_{in} and E_{out} as a function of data amount N, given the model $h()$
  - $E_{out} \geq E_{in}$ for any N
- **Small N**
  - E_{in} is small (even 0)
  - The model is likely overfitted, memorizing and generalizing poorly
  - E_{out} is large
- **Increasing N**
  - $E_{in} \uparrow$ as the model cannot fit all data
  - $E_{out} \downarrow$ as the model fits better and generalizes better
  - Generalization error $E_{out} - E_{in} \downarrow$
- $N \to \infty$
  - E_{in} reaches a maximum (irreducible error)
  - E_{out} reaches a minimum
  - $E_{out} - E_{in}$ depends on the complexity of the model

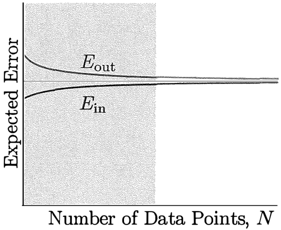




High-Bias vs High-Variance Regime

- From the learning curve you can see the two regimes:
  - **High-variance regime** for small N
    * E_{in} is small
    * Small data set $D \implies$ high dependency on D
  - More data helps reduce gap between E_{in} and E_{out}
  - **High-bias regime** for large N
    * E_{in} is large; flattens for large N
    * Best model fitted; more data won't help
    * E_{out} can be close to E_{in} (good generalization) or not

Learning curves for $\mathcal{H}_2$



Learning curves for $\mathcal{H}_{10}$